

Customer Shopping Behavior Analysis Project

Project Title

Customer Shopping Behavior Analysis Using SQL, Python (Pandas, Matplotlib), Excel & Power BI

Project Overview

This project focuses on analyzing customer shopping data using SQL, Python in Jupyter Notebook, Microsoft Excel, and Power BI. The goal is to understand customer purchasing patterns, category and product performance, seasonal sales trends, and customer demographics using data analysis and visualization techniques.

The project uses SQL for query-based analysis, Pandas for data manipulation, Matplotlib for visualization, Excel for tabular reporting, and Power BI for interactive dashboarding to extract meaningful business insights.

Objectives

- Clean and prepare the customer shopping dataset for analysis.
- Analyze overall sales performance and purchase volume.
- Understand category-wise and product-wise revenue distribution.
- Study the impact of season on customer purchasing behavior.
- Analyze customer location-wise sales performance.
- Explore customer demographics (age, gender) and their spending patterns.
- Study the impact of subscription status, discounts, and payment method on sales.
- Identify key factors driving revenue and customer retention.
- Visualize insights using graphs, charts, and an interactive dashboard.

Tools & Technologies Used

- Python 3.x
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- SQL / MySQL
- Microsoft Excel
- Power BI

Dataset Information

The dataset contains customer shopping transaction information such as:

- Customer ID
- Customer Age
- Gender
- Product Name
- Product Category
- Purchase Amount (USD)
- Customer Location
- Product Size
- Product Color
- Purchase Season
- Review Rating
- Subscription Status
- Shipping Type
- Discount Applied
- Promo Code Used
- Previous Purchases
- Payment Method
- Purchase Frequency
- Purchase Count

The dataset contains 3,900 customer transaction records across 19 fields.

Data Cleaning Steps

- Checked dataset shape, structure, and column data types.
- Generated statistical summary using `describe()`.
- Checked for missing values using `isnull().sum()` — only `Review_Rating` had missing entries (37 records, ~0.9%).
- Checked and removed duplicate records.
- Verified dataset consistency after cleaning.

Duplicate Handling

- Checked for duplicate records using `duplicated().sum()`
- Removed duplicates using `drop_duplicates()`
- Confirmed dataset has no duplicate records after cleaning

Exploratory Data Analysis (EDA)

1. Dataset Overview

- Checked dataset structure using info()
- Generated statistical summary using describe()
- Identified data types and missing values

2. Sales Overview

- Total Sales: \$233,081
- Total Purchases: 3,900
- Average Purchase Value: \$59.76
- Average Review Rating: 3.75 / 5

3. Category Analysis

- Category-wise revenue distribution (Clothing, Accessories, Footwear, Outerwear)
- Clothing generated the highest revenue at \$104,264 (44.7% of total sales)
- Outerwear generated the lowest revenue at \$18,524 (7.9% of total sales)

4. Product Analysis

- Product-wise revenue ranking using SQL and Pandas groupby
- Top-selling products: Blouse (\$10,410), Shirt (\$10,332), Dress (\$10,320), Pants (\$10,090), Jewelry (\$10,010)

5. Location Analysis

- Customer location-wise revenue distribution
- Top-performing locations: Montana, Illinois, California, Idaho, and Nevada

6. Seasonal Analysis

- Season-wise sales comparison: Fall (\$60,018), Spring (\$58,679), Winter (\$58,607), Summer (\$55,777)
- Fall recorded the highest sales; Summer recorded the lowest

7. Payment, Subscription & Discount Analysis

- Payment method-wise revenue distribution (Credit Card, PayPal, Cash, Debit Card, Venmo, Bank Transfer)
- Subscription Status vs Sales: Non-subscribers contributed \$170,436 vs \$62,645 from subscribers
- Discount Applied vs Sales: Non-discounted purchases generated \$133,670 vs \$99,411 from discounted purchases

8. Demographic Analysis

- Gender distribution and gender-wise average purchase value

- Age group-wise revenue distribution
- Male customers: 2,652 transactions, \$157,890 total sales
- Female customers: 1,248 transactions, \$75,191 total sales

Key Visualizations

The following charts were created using Python (Matplotlib), Excel, and Power BI:

1. Sales by Category

- Bar chart showing revenue distribution across product categories

2. Sales by Product

- Bar chart showing revenue generated by individual products

3. Sales by Season

- Bar chart comparing sales performance across the four seasons

4. Sales by Location

- Chart showing top customer locations by revenue

5. Payment Method Distribution

- Comparison of revenue generated across different payment methods

6. Subscription & Discount Impact

- Comparison of sales and average order value by subscription status and discount usage

7. Power BI Dashboard

- Interactive dashboard consolidating category, product, location, season, and payment views

Key Business Insights

- Overall sales performance and average purchase value were calculated.
- Clothing and Accessories together generate over 76% of total revenue.
- Outerwear underperforms compared to other categories, indicating room for growth.
- Sales are relatively stable across seasons, with Fall performing best.
- Revenue is evenly spread across payment methods, with no single method dominating.
- Non-subscribers contribute more revenue than subscribers, despite similar average order values.
- Discounted purchases do not show a higher average order value than non-discounted ones.
- Customer spend is fairly consistent across age groups, with the 45–55 age band contributing the most.

Skills Demonstrated

- Data Cleaning using Pandas
- SQL Query Writing & Database Analysis
- Exploratory Data Analysis (EDA)
- Data Visualization using Matplotlib
- Excel Reporting
- Power BI Dashboard Development
- Handling real-world retail/e-commerce datasets
- Business insight generation
- Data interpretation and storytelling

Business Value

This project helps business and marketing teams understand:

- Which product categories and products drive the most revenue
- How sales vary by season, location, and customer demographics
- Impact of subscription status and discounts on purchasing behavior
- Customer payment preferences
- Where to focus merchandising, marketing, and retention efforts

These insights can help improve inventory planning, targeted marketing, and customer retention strategies.

Conclusion

This SQL and Python-based Customer Shopping Behavior Analysis project demonstrates how data analytics techniques can be used to analyze retail transaction data and extract meaningful insights. Using SQL and Pandas for data analysis, Matplotlib for visualization, and Power BI for dashboarding, key factors affecting sales performance and customer behavior were identified. This project is suitable for entry-level Data Analyst roles and showcases strong foundational skills in SQL, Python, Excel, and Power BI-based data analysis.